

1. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030  
**Title:** New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo
2. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009  
**Title:** New binary black hole mergers in the LIGO-Virgo O3a data
3. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007  
**Title:** Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run
4. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011  
**Title:** New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO
5. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030  
**Title:** Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers

## Refereed Publications

1. Maity, K.N., Jana, S., **Venumadhav, T.**, Barsode, A., Ajith, P., (2026), Physical Review D, 113, 103522  
**Title:** Strong lensing cosmography using binary-black-hole mergers: Prospects for the near future
2. Hegade K. R., A., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), Physical Review Letters, 136, 071401  
**Title:** Relativistic and Dynamical Love Numbers
3. Cheung, M.H.Y., Wadekar, D., Mehta, A.K., Islam, T., Roulet, J., Berti, E., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2026), Physical Review D, 113, 023003  
**Title:** Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA
4. Mehta, A.K., Wadekar, D., Anantpurkar, I., Roulet, J., **Venumadhav, T.**, Islam, T., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 124023  
**Title:** Binary black hole population inference combining confident and marginal events from the IAS-HM search pipeline
5. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 112, 104039  
**Title:** Post-Newtonian theory-inspired framework for characterizing eccentricity in gravitational waveforms
6. Perera, A., Zackay, B., **Venumadhav, T.**, (2025), Astrophys. J. Suppl. Ser., 281, 4  
**Title:** A New Search Pipeline for Short Gamma-Ray Bursts in Fermi/GBM Data—A 50% Increase in the Number of Detections
7. Mushkin, J., Roulet, J., Zackay, B., **Venumadhav, T.**, Ivashtenko, O., Wadekar, D., Mehta, A.K., Zaldarriaga, M., (2025), Physical Review D, 112, 104025  
**Title:** Sampler-free gravitational wave inference using matrix multiplication
8. Jana, S., Kapadia, S.J., **Venumadhav, T.**, More, S., Ajith, P., (2025), Physical Review Letters, 135, 111402

**Title:** Probing the Nature of Dark Matter Using Strongly Lensed Gravitational Waves from Binary Black Holes

9. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 044070  
**Title:** Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms
10. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 111, L081503  
**Title:** Universal phenomenological relations between spherical harmonic modes in nonprecessing eccentric binary black hole merger waveforms
11. Mehta, A.K., Olsen, S., Wadekar, D., Roulet, J., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 111, 024049  
**Title:** New binary black hole mergers in the LIGO-Virgo O3b data
12. Jana, S., J Kapadia, S., **Venumadhav, T.**, More, S., Ajith, P., (2024), Classical and Quantum Gravity, 41, 245010  
**Title:** Strong-lensing cosmography using third-generation gravitational-wave detectors
13. Wadekar, D., **Venumadhav, T.**, Mehta, A.K., Roulet, J., Olsen, S., Mushkin, J., Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 084035  
**Title:** New approach to template banks of gravitational waves with higher harmonics: Reducing matched-filtering cost by over an order of magnitude
14. Chia, H.S., Edwards, T.D.P., Wadekar, D., Zimmerman, A., Olsen, S., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 063007  
**Title:** In pursuit of Love numbers: First templated search for compact objects with large tidal deformabilities in the LIGO-Virgo data
15. Wadekar, D., **Venumadhav, T.**, Roulet, J., Mehta, A.K., Zackay, B., Mushkin, J., Zaldarriaga, M., (2024), Physical Review D, 110, 044063  
**Title:** New search pipeline for gravitational waves with higher-order modes using mode-by-mode filtering
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**Title:** Fast marginalization algorithm for optimizing gravitational wave detection, parameter estimation, and sky localization
17. Roulet, J., **Venumadhav, T.**, (2024), Annu. Rev. Nucl. Part. Sci., 74, 207  
**Title:** Inferring Binary Properties from Gravitational-Wave Signals
18. Yu, H., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2023), Physical Review D, 108, 064059  
**Title:** Accurate and efficient waveform model for precessing binary black holes
19. Jana, S., Kapadia, S.J., **Venumadhav, T.**, Ajith, P., (2023), Physical Review Letters, 130, 261401  
**Title:** Cosmography Using Strongly Lensed Gravitational Waves from Binary Black Holes
20. Yu, H., Weinberg, N.N., Arras, P., Kwon, J., **Venumadhav, T.**, (2023), Mon. Not. R. Astron. Soc., 519, 4325  
**Title:** Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals
21. Roulet, J., Olsen, S., Mushkin, J., Islam, T., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 123015  
**Title:** Removing degeneracy and multimodality in gravitational wave source parameters

22. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009  
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**Title:** Signs of higher multipoles and orbital precession in GW151226
24. Olsen, S., Roulet, J., Chia, H.S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083036  
**Title:** Mapping the likelihood of GW190521 with diverse mass and spin priors
25. Roulet, J., Chia, H.S., Olsen, S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083010  
**Title:** Distribution of effective spins and masses of binary black holes from the LIGO and Virgo O1-O3a observing runs
26. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030  
**Title:** Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers
27. Zackay, B., **Venumadhav, T.**, Roulet, J., Dai, L., Zaldarriaga, M., (2021), Physical Review D, 104, 063034  
**Title:** Detecting gravitational waves in data with non-stationary and non-Gaussian noise
28. Roulet, J., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 123022  
**Title:** Binary black hole mergers from LIGO/Virgo O1 and O2: Population inference combining confident and marginal events
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32. Samsing, J., **Venumadhav, T.**, Dai, L., Martinez, I., Batta, A., Lopez, M., Ramirez-Ruiz, E., Kremer, K., (2019), Physical Review D, 100, 043009  
**Title:** Probing the black hole merger history in clusters using stellar tidal disruptions
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**Title:** Highly Magnified Stars in Lensing Clusters: New Evidence in a Galaxy Lensed by MACS J0416.1-2403
34. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007  
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**Title:** New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO
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**Title:** Template bank for compact binary coalescence searches in gravitational wave data: A general geometric placement algorithm
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**Title:** Probing Dark Matter Subhalos in Galaxy Clusters Using Highly Magnified Stars
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**Title:** Heating of the intergalactic medium by the cosmic microwave background during cosmic dawn
39. Kaurov, A.A., **Venumadhav, T.**, Dai, L., Zaldarriaga, M., (2018), Astrophysical Journal, 864, L15  
**Title:** Implication of the Shape of the EDGES Signal for the 21 cm Power Spectrum
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**Title:** Microlensing of Extremely Magnified Stars near Caustics of Galaxy Clusters
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**Title:** Effect of lensing magnification on the apparent distribution of black hole mergers
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**Title:** Stability of small-scale baryon perturbations during cosmological recombination
48. **Venumadhav, T.**, Zimmerman, A., Hirata, C.M., (2014), Astrophysical Journal, 781, 23  
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49. Venumadhav, T., Haque, M., Moessner, R., (2010), Physical Review B, 81, 054305  
**Title:** Finite-rate quenches of site bias in the Bose-Hubbard dimer

## Preprints

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**Title:** The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response

2. Islam, T., Wadekar, D., **Venumadhav, T.**, Zaldarriaga, M., Mehta, A.K., Roulet, J., Zackay, B., (2026), arXiv:2605.11280  
**Title:** Discovery of Interpretable Surrogates via Agentic AI: Application to Gravitational Waves
3. Perera, A., Zackay, B., **Venumadhav, T.**, (2026), arXiv:2605.31554  
**Title:** Expanding the Population of Short Gamma-Ray Transients with a Coherent Fermi/GBM Search. A 13-year catalog of short GRBs
4. Islam, T., **Venumadhav, T.**, Wadekar, D., Mehta, A.K., Roulet, J., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2604.07388  
**Title:** GW190711\_030756 and GW200114\_020818: astrophysical interpretation of two asymmetric binary black hole mergers in the IAS catalog
5. Zhou, Z., Wadekar, D., Roulet, J., Ivashtenko, O., **Venumadhav, T.**, Islam, T., Mehta, A.K., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2603.05784  
**Title:** Searching for precessing binary systems with mode-by-mode filtering and marginalization
6. Islam, T., **Venumadhav, T.**, Wadekar, D., (2026), arXiv:2601.18986  
**Title:** Progenitor of the recoiling super-massive black hole RBH-1 identified using HST/JWST imaging
7. Islam, T., **Venumadhav, T.**, Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2509.20556  
**Title:** Data-driven extraction and phenomenology of eccentric harmonics in eccentric spinning binary black hole mergers
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9. Schiff, J., **Venumadhav, T.**, (2025), arXiv:2506.16517  
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**Title:** Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of  $g$ -modes in merging neutron star binaries
12. Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., Islam, T., (2025), arXiv:2501.17939  
**Title:** Significant increase in sensitive volume of a gravitational wave search upon including higher harmonics
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15. Islam, T., Roulet, J., **Venumadhav, T.**, (2022), arXiv:2210.16278  
**Title:** Factorized Parameter Estimation for Real-Time Gravitational Wave Inference
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**Title:** Search for Lensed Gravitational Waves Including Morse Phase Information: An Intriguing Candidate in O2
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**Title:** Identifying strongly lensed gravitational wave signals from binary black hole mergers
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**Title:** On the waveforms of gravitationally lensed gravitational waves

## $n^{\text{th}}$ Author Papers

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1. Raaijmakers, G., et. al., (2021), Astrophysical Journal, 922, 269  
**Title:** The Challenges Ahead for Multimessenger Analyses of Gravitational Waves and Kilonova: A Case Study on GW190425
2. Weltman, A., et. al., (2020), Publ. Astron. Soc. Aust., 37, e002  
**Title:** Fundamental physics with the Square Kilometre Array
3. Doré, O., et. al., (2014), arXiv:1412.4872  
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